

```
In [2]: # This Python 3 environment comes with many helpful analytics libraries installed
# It is defined by the kaggle/python docker image: https://github.com/kaggle/docker
# For example, here's several helpful packages to load in
import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import seaborn as sns
import matplotlib.pyplot as plt
# Input data files are available in the "../input/" directory.
# For example, running this (by clicking run or pressing Shift+Enter) will list all
```

2.Load the file

```
In [6]: income_df = pd.read_csv(r"C:\Users\ANIL\Desktop\Inc_Exp_Data.csv")
```

```
In [33]: income_df.head()
```

```
Out[33]:
```

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income
0	5000	8000	3	2000	
1	6000	7000	2	3000	
2	10000	4500	2	0	
3	10000	2000	1	0	
4	12500	12000	2	3000	

3.Analyze the data

```
In [34]: income_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Mthly_HH_Income                       50 non-null     int64
1   Mthly_HH_Expense                     50 non-null     int64
2   No_of_Fly_Members                    50 non-null     int64
3   Emi_or_Rent_Amt                      50 non-null     int64
4   Annual_HH_Income                     50 non-null     int64
5   Highest_Qualified_Member              50 non-null     object
6   No_of_Earning_Members                 50 non-null     int64
dtypes: int64(6), object(1)
memory usage: 2.9+ KB
```

```
In [9]: income_df.shape
```

Out[9]: (50, 7)

In [12]: `income_df.describe().T`

Out[12]:

	count	mean	std	min	25%	50%	
Mthly_HH_Income	50.0	41558.00	26097.908979	5000.0	23550.0	35000.0	50
Mthly_HH_Expense	50.0	18818.00	12090.216824	2000.0	10000.0	15500.0	25
No_of_Fly_Members	50.0	4.06	1.517382	1.0	3.0	4.0	
Emi_or_Rent_Amt	50.0	3060.00	6241.434948	0.0	0.0	0.0	3
Annual_HH_Income	50.0	490019.04	320135.792123	64200.0	258750.0	447420.0	594
No_of_Earning_Members	50.0	1.46	0.734291	1.0	1.0	1.0	

◀  ▶

In [13]: `income_df.isna().any()`

Out[13]:

Mthly_HH_Income	False
Mthly_HH_Expense	False
No_of_Fly_Members	False
Emi_or_Rent_Amt	False
Annual_HH_Income	False
Highest_Qualified_Member	False
No_of_Earning_Members	False

dtype: bool

4.What is the Mean Expense of a Household?

In [14]: `income_df["Mthly_HH_Expense"].mean()`

Out[14]: 18818.0

5. What is the Median Household Expense?

In [15]: `income_df["Mthly_HH_Expense"].median()`

Out[15]: 15500.0

6.What is the Monthly Expense for most of the Household?

```
In [19]: mth_exp_tmp = pd.crosstab(index=income_df["Mthly_HH_Expense"], columns="count")
mth_exp_tmp.reset_index(inplace=True)
mth_exp_tmp[mth_exp_tmp['count'] == income_df.Mthly_HH_Expense.value_counts().max()]
```

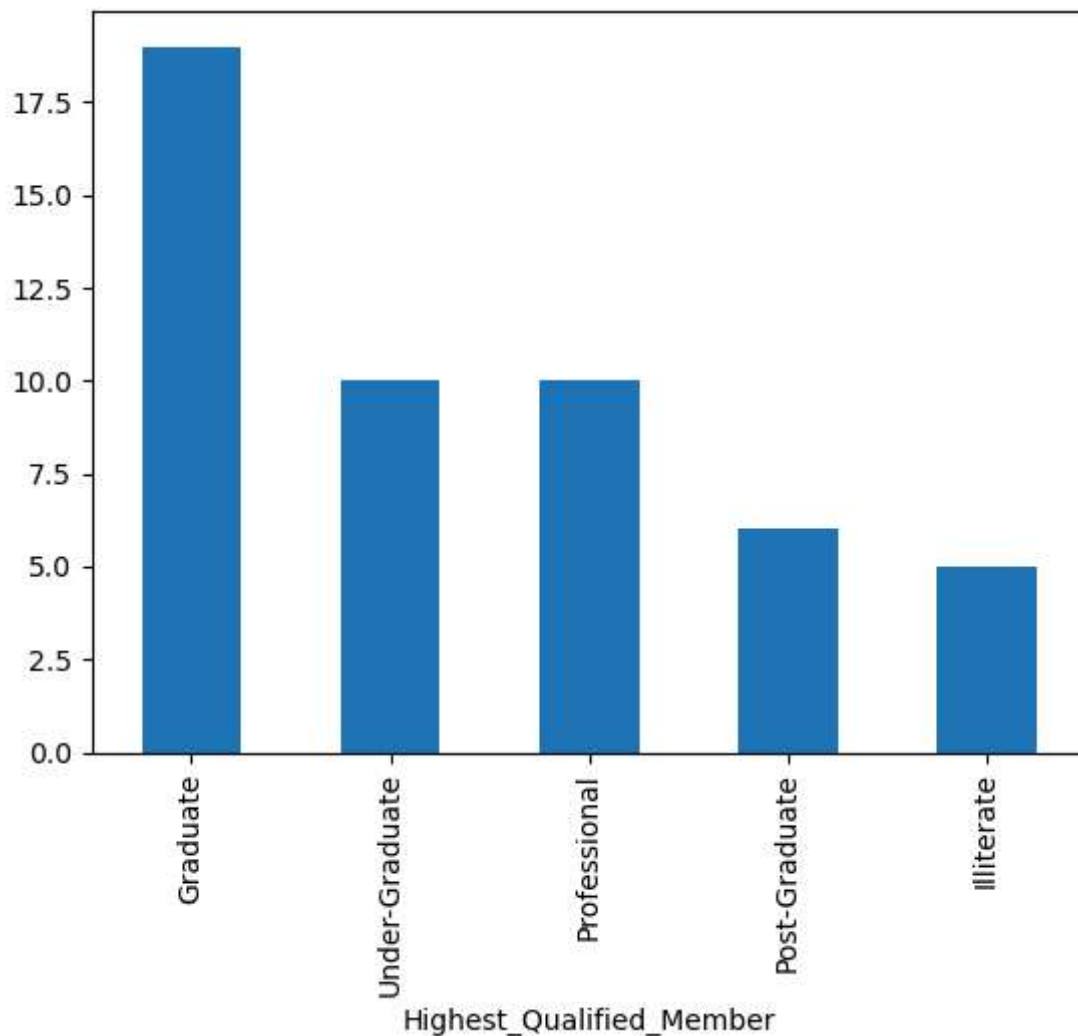
```
Out[19]:
```

col_0	Mthly_HH_Expense	count
18	25000	8

7.plot the Histogram to count the Highest qualified member

```
In [20]: income_df["Highest_Qualified_Member"].value_counts().plot(kind="bar")
```

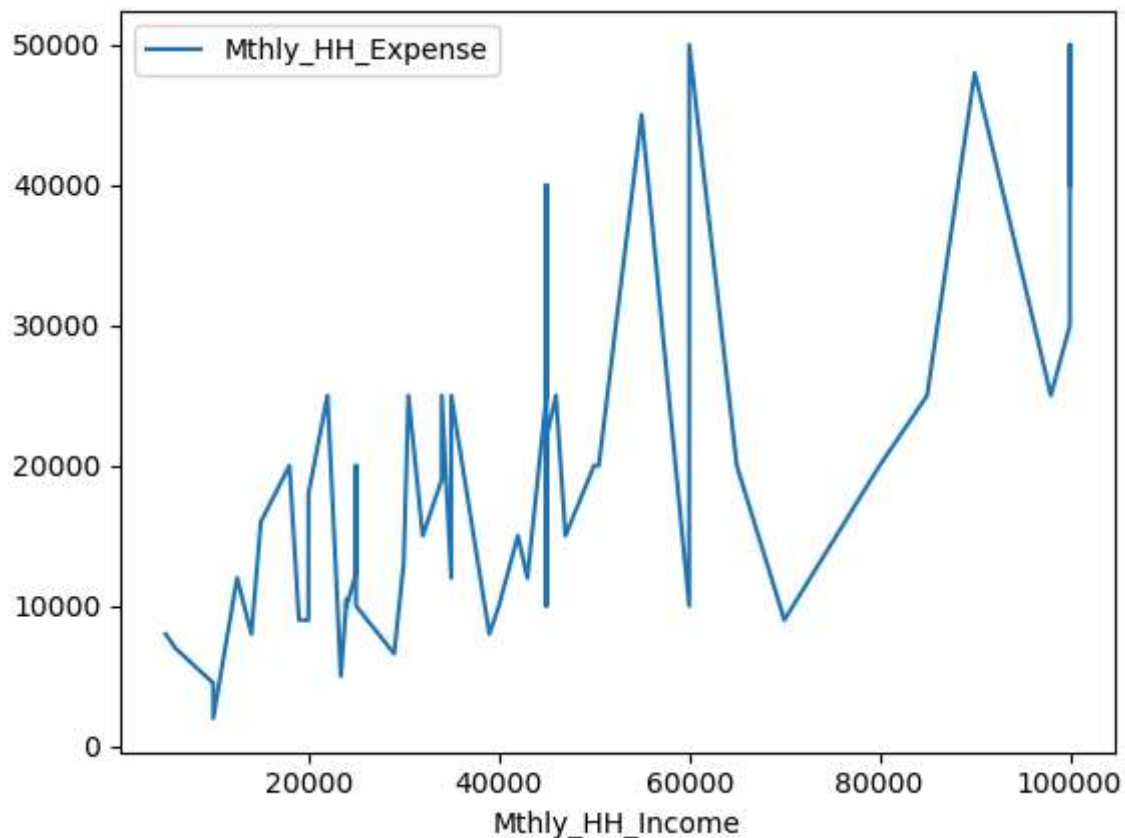
```
Out[20]: <Axes: xlabel='Highest_Qualified_Member'>
```



8.Calculate IQR(difference)btween 75% and 25% quartile

```
In [25]: income_df.plot(x="Mthly_HH_Income", y="Mthly_HH_Expense")
IQR=income_df["Mthly_HH_Expense"].quantile(0.75)-income_df["Mthly_HH_Expense"].quan
IQR
```

Out[25]: 15000.0



9. Calculate Standard Deviation For first 4 columns.

```
In [26]: pd.DataFrame(income_df.iloc[:,0:5].std().to_frame()).T
```

Out[26]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income
0	26097.908979	12090.216824	1.517382	6241.434948	32013.561113



10. Calculate Variance for first 3 columns.

```
In [27]: pd.DataFrame(income_df.iloc[:,0:4].var().to_frame()).T
```

Out[27]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt
0	6.811009e+08	1.461733e+08	2.302449	3.895551e+07

11. Calculate the count of Highest qualified member.

In [28]: `income_df["Highest_Qualified_Member"].value_counts().to_frame().T`

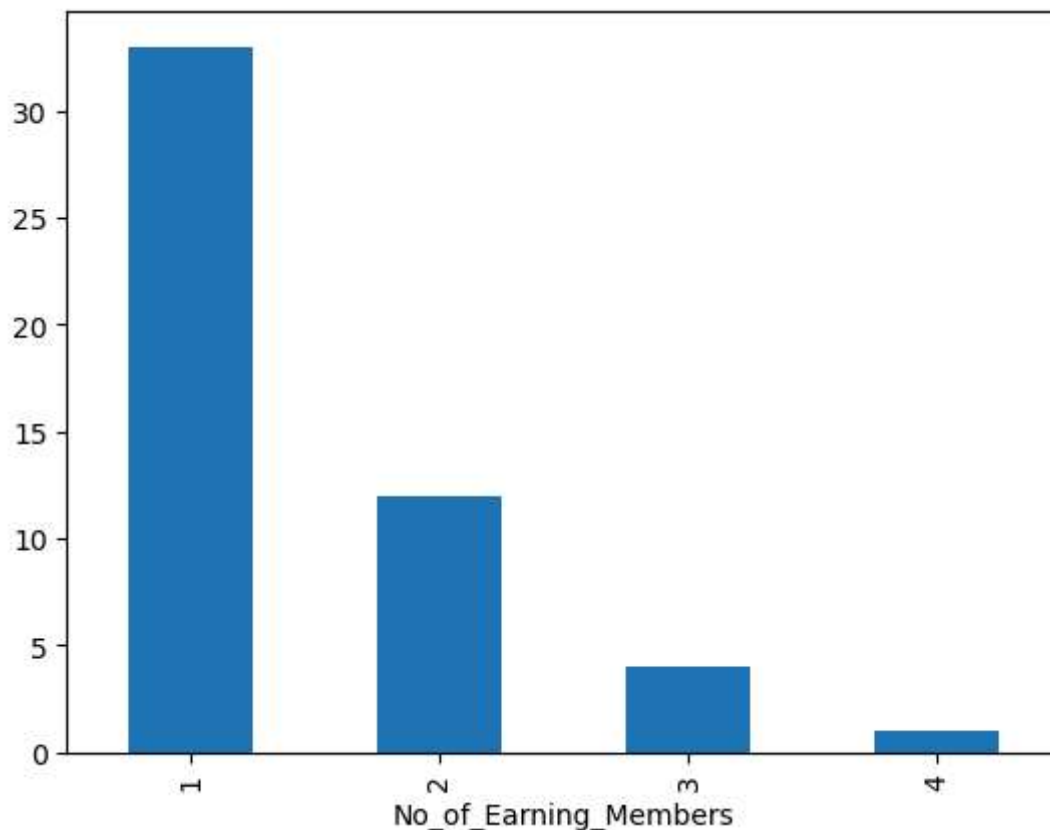
Out[28]:

Highest_Qualified_Member	Graduate	Under-Graduate	Professional	Post-Graduate	Illiterate
count	19	10	10	6	5

12. Plot the Histogram to Count the No_of_Earning_Members

In [29]: `income_df["No_of_Earning_Members"].value_counts().plot(kind="bar")`

Out[29]: <Axes: xlabel='No_of_Earning_Members'>



13 . Suppose you have option to invest in Stock A or Stock B. The stocks . have different expected returns and standard deviation. The expected return of Stock A is 15% and Stock B is 10% Standard Deviation of the returns of these stocks is 10% and 5% respectively.

In [30]: *# Here we need to calculate the coeff of variation*

```
coeff_of_var_StockA=10/15  
print(coeff_of_var_StockA)  
coeff_of_var_StockB=5/10  
print(coeff_of_var_StockB)
```

0.6666666666666666

0.5

In []:

In []:

In []: